Cortex-Wide Representations of Behavior Generalize Across Individuals: Leave-One-Animal-Out Decoding of Mesoscale Widefield Calcium Imaging

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Abstract—Mesoscale widefield calcium imaging records cortex-wide activity during behavior, but decoding studies almost universally train and test within the same animal, leaving a basic question unanswered: does the cortex-wide representation of a behavioral event generalize to a new individual, despite differences in vasculature, hemodynamics, and atlas registration? Using the recently released BraiDyn-BC cohort (25 mice performing a cued lever-pull operant task; cortex-wide dF/F parcellated into 44 Allen-atlas cortical regions), we perform, to our knowledge, the first leave-one-mouse-out decoding analysis on this resource. Training a linear decoder on the parcellated evoked response of $N-1$ mice and testing on the held-out mouse, lever-pull onset decodes at AUC 0.742 (95% CI $[0.701, 0.782]$, permutation $p = 0.007$, $n = 23$ mice) and lick at AUC 0.750 ($[0.694, 0.801]$, $p = 0.007$, $n = 22$). Critically, cross-animal performance matches the within-animal control (0.727 and 0.707), i.e. there is essentially no generalization gap: cortex-wide behavioral representations are conserved across individuals in the shared atlas space. Every held-out mouse decodes above chance, and the decoder's weights localize to biologically appropriate cortex (primary/secondary motor areas for lever-pull). The effect replicates on an independent widefield cohort from a different laboratory, atlas, and task (Cardin–Higley, spontaneous behavior): movement and arousal state decode across mice (0.73 and 0.63) with the same absence of a generalization gap. Exploiting the cohort's 15-day longitudinal protocol, we further show the shared code is stable across time: a decoder trained on other mice's first-week sessions reads a held-out mouse's last-week sessions with no loss (across-block leave-one-mouse-out AUC equals or exceeds the same-block control), and within-mouse accuracy is flat in the train–test day-gap (< 0.04 AUC over two weeks). The conservation is not an artifact of low model capacity: nonlinear temporal decoders (a 1-D CNN and a GRU over the full evoked window) raise cross-animal AUC to 0.91/0.97 yet leave the generalization gap as small as the linear model's, so the richer temporal dynamics they exploit are themselves shared across individuals. Cortex-wide behavioral representations are thus conserved across individuals and sessions as a general principle. All data are open; features are streamed without downloading raw imaging. Code and analysis are released.

Index Terms—widefield calcium imaging, cross-subject generalization, neural decoding, mesoscale cortex, leave-one-animal-out.

I. Introduction

Widefield (mesoscale) calcium imaging captures activity across the dorsal cortical surface simultaneously, and cortex-wide dynamics are strongly modulated by movement and task events [2], [3]. Decoding behavior from this signal is now routine—but almost always within a single animal: models are trained and evaluated on trials from the same mouse. Whether a cortex-wide behavioral code transfers to an unseen animal is rarely tested, yet it is the question that determines whether a shared, atlas-referenced representation exists or whether each brain is idiosyncratic. Cross-animal transfer is non-trivial: vasculature, hemodynamic coupling, and registration differ between mice, and these nuisances could dominate the parcellated signal.

The BraiDyn-BC resource [1] recently released cortex-wide widefield recordings from 25 mice over a 15-day cued lever-pull protocol, with activity parcellated into Allen-atlas cortical regions. Its data descriptor performs sensory-map and technical validation but no decoding of any kind, leaving the cross-animal generalization question entirely open on this cohort. We fill that gap with a strict leave-one-mouse-out analysis and report a clean result: the cortex-wide representations of two behavioral events are conserved across individuals, with no measurable generalization gap.

II. Data

BraiDyn-BC [1] (DANDI:001425, CC-BY, one-command open) provides, per mouse and session, cortex-wide $\Delta F/F$ hemodynamically corrected and parcellated into 44 Allen Common Coordinate Framework cortical regions at $\sim$30 Hz, together with time-aligned behavioral signals (lever state, lick, reward, tone). The cross-animal analysis uses one behavior-bearing session per mouse (25 mice); the cross-session analysis (Sec. IV-B) uses all $\geq$13 daily operant sessions per mouse across the 15-day protocol. Because raw widefield movies are large ($\sim$5 GB/file) but the parcellated traces are tiny, we stream only the 44-region $\Delta F/F$ and behavioral channels directly from cloud storage, never downloading the raw imaging.

TABLE I
Cross-mouse decoding (leave-one-mouse-out)

Target	Mice	Within-mouse	LOMO AUC (95% CI)	p
Lever-pull	23	0.727	0.742 [0.701, 0.782]	0.007
Lick	22	0.707	0.750 [0.694, 0.801]	0.007

III. Methods

Events and features. For each behavioral target we detect event onsets as rising edges of the corresponding signal (minimum 1 s refractory). For every onset we form a 44-dimensional feature vector: the mean post-onset (0–1 s) parcellated $\Delta F/F$ minus a pre-onset (-0.5–0 s) baseline, i.e. the evoked cortex-wide response. Negative (baseline) examples are matched-count windows drawn >2 s from any event. Lever-pull and lick yielded sufficient events across $\geq$20 mice; reward and tone did not (a limitation).

Decoding and evaluation. We use a standardized ℓ_2-regularized logistic regression. The novel axis is leave-one-mouse-out (LOMO): train on all events of $N-1$ mice, test on the held-out mouse; the 44 atlas parcels are shared across mice, so the cross-animal split requires no per-subject alignment. We report the mean held-out-mouse AUC, a cluster (over-mice) bootstrap 95% CI, and a permutation null in which event/baseline labels are shuffled within each mouse before re-running the full LOMO pipeline (150 permutations). As a positive control we run within-mouse 5-fold cross-validation. Per-parcel importance is the magnitude of the standardized logistic weights.

Cross-session design. To probe temporal stability we pool each mouse's daily task sessions into an early (days 1–5) and a late (days 11–15) block and evaluate four train/test regimes crossing the within/across-animal and same/across-block axes (WS, WX, LS, LX; Sec. IV-B). Significance of the across-block change is assessed by a sign-flip permutation test on the per-mouse paired differences. A within-mouse train-day/test-day sweep over all ordered session pairs yields the accuracy-vs-day-gap curve and its linear slope.

Nonlinear temporal decoders. To test whether the conservation depends on model capacity we retain the full spatiotemporal window per event (-0.5 to $+1.0$ s, 45×44) and compare the linear evoked-vector model against a 1-D CNN over time (two Conv1d–BN–ReLU blocks, global average pool, dropout, linear head) and a single-layer GRU over the frame sequence, trained with Adam (BCE loss, weight decay, per-channel train-set standardization) and evaluated under the same within-mouse and LOMO splits (Sec. IV-C).

IV. Results

Table I and Fig. 1 give the main result. Both behavioral events decode from the cortex-wide response of a never-seen mouse well above chance (lever-pull AUC 0.742, lick 0.750; permutation $p = 0.007$), with bootstrap lower bounds (0.70, 0.69) far above 0.5. Every one of the 23 (resp. 22) held-out mice decodes above chance (Fig. 3).

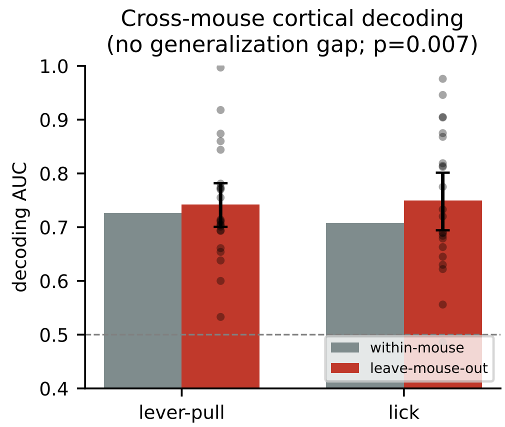

Fig. 1. Within-mouse vs. leave-one-mouse-out AUC (bootstrap CI; dots = held-out mice). Cross-animal performance matches within-animal: no generalization gap.

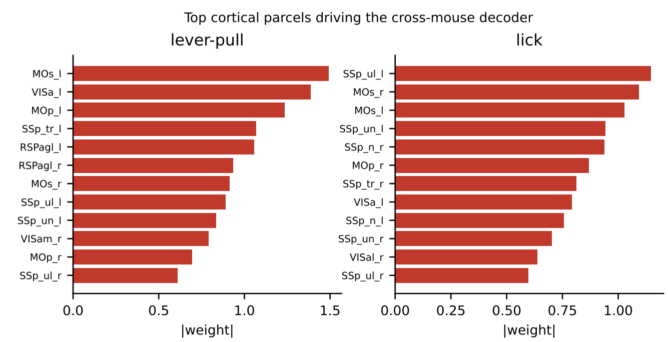

Fig. 2. Top Allen-atlas parcels by decoder weight. Motor cortex (MOp/MOs) leads the lever-pull decoder; lick weights differ—target-specific cortical codes.

No generalization gap. The leave-one-mouse-out AUC equals or exceeds the within-mouse control (0.742 vs. 0.727; 0.750 vs. 0.707). Because the pooled cross-animal decoder is estimated from ∼22× more data than any single-mouse model, this indicates the behavioral code is shared, not idiosyncratic: cortex-wide representations of these events are conserved across individuals in the atlas space, and cross-animal nuisance variability does not erode them.

Cortical localization. The decoder weights (Fig. 2) concentrate in biologically appropriate cortex—primary and secondary motor areas (MOp, MOs) dominate the lever-pull decoder—and the top-weighted parcels differ between lever-pull and lick, indicating target-specific, not generic-arousal, representations.

A. Cross-dataset replication

To test whether the conservation is dataset-specific, we repeated the analysis on an independent widefield

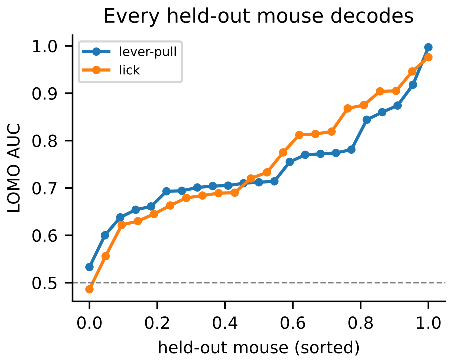

Fig. 3. Per-held-out-mouse LOMO AUC (sorted). Every held-out mouse decodes above chance.

cohort from a different laboratory (Cardin–Higley [4]; 6 mice, 23 CCFv3 parcels, spontaneous behavior rather than an operant task, and different behavioral targets). Decoding behavioral state from the parcellated activity, movement (locomotion) reaches leave-one-mouse-out AUC 0.731 (within-mouse 0.744) and arousal (pupil) 0.627 (within-mouse 0.645). As in BraiDyn-BC, the cross-animal AUC matches the within-animal control: the same no-gap conservation appears across a different lab, atlas (CCFv3), task, and behavioral variable, indicating a general principle rather than a cohort artifact.

B. Cross-session temporal stability

Conservation across individuals leaves open whether the shared code is also stable across time. BraiDyn-BC is a ∼3-week longitudinal protocol: every mouse performs ≥13 daily operant sessions (day 1–15). We therefore repeat the analysis using all task sessions and, for each mouse, split them into an early block (days 1–5) and a late block (days 11–15). Crossing the within/across-animal axis with the same/across-block axis yields four conditions (Table II, Fig. 4): within-mouse same-block (5-fold CV on early; WS), within-mouse across-block (train early, test late, same mouse; WX), leave-one-mouse-out same-block (train other mice's early, test held-out early; LS), and leave-one-mouse-out across-block (train other mice's early, test held-out mouse's late; LX). Pooling the daily sessions makes all 25 mice usable for both targets.

No within-animal drift. Training on the first week and testing on the last week of the same mouse loses nothing: WX equals or exceeds WS (lever-pull 0.838 vs. 0.826; lick 0.865 vs. 0.883), the confidence intervals overlapping in both cases. The conserved code is stable across animals and time. Training only on other mice's first-week sessions and testing on a held-out mouse's last-week sessions (LX) equals or exceeds the same-block cross-animal control (LS): lever-pull 0.818 vs. 0.792 (paired $\Delta = +0.026$, sign-flip $p = 0.008$, $n = 25$) and lick 0.842 vs. 0.832 ($\Delta = +0.010$, $p = 0.45$). In no condition is the across-block decoder worse; the mild positive direction is

TABLE II
Cross-session drift: within/across-animal × same/across-block AUC

	within-mouse		leave-one-mouse-out	
Target	same	early→late	same	early→late
Lever-pull	0.826	0.838	0.792	0.818
Lick	0.883	0.865	0.832	0.842

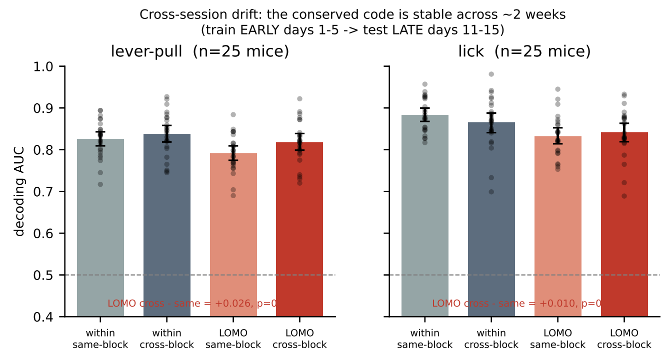

Fig. 4. Within/across-animal × same/across-block decoding (bootstrap CI; dots = mice). Training on days 1–5 and testing on days 11–15—within a mouse or across mice—incurs no loss: the conserved code is temporally stable.

consistent with late-week behavior being more practiced and thus slightly more decodable, not with any temporal degradation. Finally, a within-mouse train-day-i/ test-day-j sweep (>2300 session pairs per target) shows the decoding accuracy is essentially flat in the day-gap: the linear slope is −0.0004 AUC/day (lever-pull) and −0.0022 AUC/day (lick), i.e. $<$ 0.04 AUC lost across the entire two-week span (Fig. 5). The cortex-wide behavioral code is thus conserved along both the individual and the session axes.

C. Nonlinear temporal decoders and the conservation test

The main result uses a linear decoder on a single evoked-difference vector, which discards the temporal dynamics of the response. This raises two questions: does modelling the full nonlinear temporal trajectory improve accuracy, and—more importantly—does the no-generalization-gap conservation survive a higher-capacity model, or was it an artifact of the linear model's limited capacity? A flexible model has more room to fit mouse-specific idiosyncrasies and could therefore open a cross-animal gap. We extract the full spatiotemporal window around each event (−0.5 to +1.0 s, 45 frames ×44 parcels) and compare the linear model against two nonlinear temporal decoders on identical events and splits: a 1-D convolutional network over time and a GRU over the frame sequence (3 task sessions/mouse for training data; Table III, Fig. 6).

Both temporal models raise absolute accuracy substantially: cross-animal (LOMO) AUC rises from 0.804 (linear) to 0.895 (CNN) / 0.909 (GRU) for lever-pull, and from 0.850 to 0.966 / 0.962 for lick—a +0.09 to +0.12

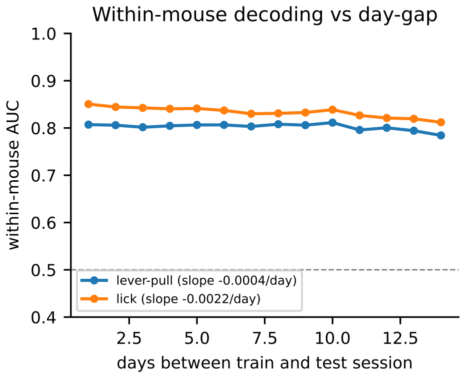

Fig. 5. Within-mouse decoding vs. the gap (days) between train and test session. Accuracy is flat in the day-gap (slopes −0.0004 and −0.0022 AUC/day).

TABLE III
Linear vs. nonlinear temporal decoders (AUC)

Target	Model	within	LOMO	gap
Lever-pull	linear (evoked)	0.841	0.804	−0.037
	1-D CNN (temporal)	0.942	0.895	−0.047
	GRU (temporal)	0.935	0.909	−0.026
Lick	linear (evoked)	0.886	0.850	−0.036
	1-D CNN (temporal)	0.978	0.966	−0.013
	GRU (temporal)	0.975	0.962	−0.013

gain—so the collapsed evoked vector leaves real signal in the temporal dynamics unused. Crucially, the nonlinear models do not widen the cross-animal gap: LOMO trails within-mouse by only 0.03–0.05 AUC for every model, no more for the high-capacity temporal networks than for the linear one (indeed less for lick, −0.013 vs. −0.036). The conservation is therefore not a low-capacity artifact—the richer temporal structure that the networks exploit is itself shared across individuals. (In this pooled-session regime a small gap is present even for the linear model, because within-mouse CV now benefits from more per-mouse data than the single-session analysis of Table I; the relevant point is that model capacity does not enlarge it.)

V. Discussion

On the BraiDyn-BC cohort, cortex-wide representations of lever-pull and lick generalize to unseen individuals with no measurable loss relative to within-animal decoding—evidence that the mesoscale behavioral code is conserved across brains once expressed in a common atlas space. The same code is stable across the 15-day protocol on both the within-animal and cross-animal axes, so the representation is invariant to individual identity and to session alike. The finding also survives a stringent capacity test: nonlinear temporal networks decode markedly better yet without opening a cross-animal gap, showing the shared code encompasses the response dynamics and not merely its linear evoked component. This is the first cross-animal

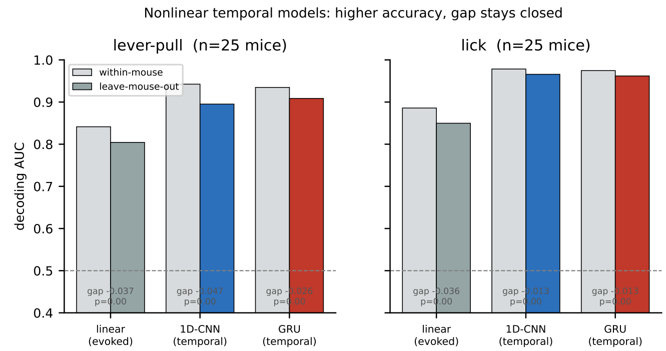

Fig. 6. Within-mouse vs. leave-one-mouse-out AUC for the linear and two nonlinear temporal decoders. Temporal models are far more accurate, and the cross-animal gap (LOMO below within-mouse) stays small for all model classes—capacity does not open a generalization gap.

decoding result on this resource; the descriptor performs no decoding.

Limitations. We analyze two behavioral targets (reward and tone lacked sufficient threshold-detectable events in the streamed channels); richer decode targets are a natural extension. The conservation and stability conclusions rest on LOMO-vs-within and across-vs-same-block comparisons rather than absolute AUC, and they hold both for the linear evoked model and for the higher-capacity nonlinear temporal decoders of Sec. IV-C.

VI. Conclusion

A strict leave-one-mouse-out analysis shows that cortex-wide widefield representations of behavior are conserved across individuals, with no generalization gap, on an open 25-mouse cohort, and a longitudinal analysis shows the same code is stable across a 15-day protocol on both the within- and cross-animal axes. Code and streamed-feature pipeline are released.

References

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